It is a principle that shines impartially on the just and the unjust that once you have a point of view all history will back you up.

Van Wyck Brooks, *America's Coming-of-Age* (New York, 1915), 20.

The rage for wanting to conclude is one of the most deadly and most fruitless manias to befall humanity. Each religion and each philosophy has pretended to have God to itself, to measure the infinite, and to know the recipe for happiness. What arrogance and what nonsense! I see, to the contrary, that the greatest geniuses and the greatest works have never concluded.

Gustave Flaubert, *Correspondance* (Paris, 1929), vol. v, 111.

[The anthropological idea of "culture" is] so diffuse and all-embracing as to seem like an all-seasons explanation for anything human beings might contrive to do, imagine, say, be, or believe. . . . We were condemned, it seemed, to working with a logic and a language in which concept, cause, form, and outcome had the same name.

Clifford Geertz, *Available Light* (Princeton, 2000), 12–13.

Words strain,
Crack and sometimes break, under the burden,
Under the tension, slip, slide, perish,
Decay with imprecision, will not stay in place,
Will not stay still.

T. S. Eliot, "Burnt Norton," *Four Quartets* (London, 1943).

First you establish the traditional "two views" of the question. You then put forward a common-sensical justification of the one, only to refute it by the other. Finally, you send them both packing by the use of a third interpretation, in which both the others are shown to be equally unsatisfactory. Certain verbal maneuvers enable you to line up the traditional "antitheses" as complementary aspects of a single reality: form and substance, content and container, appearance and reality, essence and existence, continuity and discontinuity, and so on. Before long the exercise becomes the merest verbalizing, reflection gives place to a kind of superior punning, and the "accomplished philosopher" may be recognized by the ingenuity with which he makes ever-bolder play with assonance, ambiguity, and the use of those words which sound alike and yet bear quite different meanings.

Claude Lévi-Strauss, *Tristes Tropiques* (Paris, 1955; London, 1961), 54.

Corruption in Evidence Presentations: Effects Without Causes, Cherry-Picking, Overreaching, Chartjunk, and the Rage to Conclude

MAKING a presentation is a moral act as well as an intellectual activity. The use of corrupt manipulations and blatant rhetorical ploys in a report or presentation—outright lying, flagwaving, personal attacks, setting up phony alternatives, misdirection, jargon-mongering, evading key issues, feigning disinterested objectivity, willful misunderstanding of other points of view—suggests that the presenter lacks both credibility and evidence. To maintain standards of quality, relevance, and integrity for evidence, consumers of presentations should insist that presenters be held intellectually and ethically responsible for what they show and tell. Thus *consuming* a presentation is also an intellectual and a moral activity.

THESE 2 chapters describe widely used presentation methods that are enemies of the truth, that corrupt reasoning, that produce unbeautiful anti-evidence. Rather than blatant, the methods described are subtle, the more dangerous for being so. These maneuvers often distort evidence, deceive the audience, and exploit the bond of trust necessary for reliable and intelligent communication.

CORRUPT maneuvers, blatant or subtle, are epidemic in political speeches, marketing, internet rants, and PowerPoint pitches. Less often, perhaps, corruptions of reasoning and evidence show up in serious presentations: our examples in these 2 chapters include strategic plans in business, a report of a presidential commission, economics, engineering analysis during a crisis, certain data analysis techniques, public health, medical research, and social science.

DESPITE the threat of corruption, a consumer of presentations should try to be hopeful and curious, avoid premature skepticism, and maintain an open mind but not an empty head. After all, many presentations are not corrupt. Furthermore, a presenter engaging in corrupt maneuvers might be reporting what eventually turn out to be accurate and truthful conclusions. A particular danger, then, of corrupt maneuvers is not only that they enable lying but also that they place the truth in disrepute. From scientific reports to political speeches, few things are more appalling than listening to inept and specious arguments made by one's allies.

Effects Without Causes and The Evasion of Responsibility

The *Report* of the 9/11 Commission describes major lapses in security and missed opportunities that may have allowed the attacks to take place:

> [Despite the CIA's numerous warnings, America's] domestic agencies never mobilized in response to the threat. They did not have direction, and did not have a plan to institute. The borders were not hardened. Transportation systems were not fortified. Electronic surveillance was not targeted against a domestic threat. State and local law enforcement were not marshaled to augment the FBI's efforts. The public was not warned.[1]

[1] *The 9/11 Commission Report: Final Report of the National Commission on Terrorist Attacks Upon the United States* (New York, 2004), 265.

Here and elsewhere in the *Report*, conspicuously absent is the agent of inaction. Above, 5 verbs are passive. The 3 active verbs take the utterly vague subject "domestic agencies." Exactly *who* did not make a plan, *who* did not follow up, *who* failed to warn the public? "These things that were not done must have been not done by somebody, and the somebodies reporting to him."[2] By means of the passive voice, the 9/11 Commission evades attributing responsibility for security lapses. Of course, if the passive didn't exist, they would have done it some other way.

[2] Thomas Powers, "How Bush Got It Wrong," *The New York Review of Books*, 60 (September 23, 2004), 87.

Although often a useful writing technique, passive verbs also advance *effects without causes*, an immaculate conception. To speak of ends without means, agency without agents, actions without actors is contrary to clear thinking. If the issues at hand involve responsibilities or decisions or plans, causal reasoning is necessary. The logic of decisions is "If we do such-and-such [cause], then we hope this-and-that will happen [desired effect]." And the logic of responsibility is the logic of the active voice: *someone did* or *did not do* something. Alert audiences should watch out for causality from nowhere and its sometime assistant, the passive voice.

The technique of evasion by passive voice is well-known and widely noted—and yet some reviewers of *The 9/11 Commission Report* failed to detect its overt evasive deflections of responsibility. What is obvious, and perhaps even tiresome to the sophisticated, is not so obvious in serious action. Why should we fail to be vigilant and rigorous about the quality of evidence and its presentation just because a report is part of a public dialogue, or is meant for the news media, or is from the government, or concerns an important matter?

WHAT the passive voice can be for verbal reasoning, certain statistical methods are for data reasoning: anti-causal, a jumble of effects without causes. In particular, the methodologies of *data mining*, *factor analysis*, and *multidimensional scaling* crunch and grind vast data matrices down into smaller lumps. These methodologies, however, fail to testify about causal models. Such techniques are perhaps useful for those who have lots of data but no ideas, to those otherwise clueless about causality. To be relevant for decisions and actions, whatever emerges from data crunching must somehow turn into real evidence about causal processes.

LIKE the passive voice, the *bullet-list format* collaborates with evasive presenters to promote effects without causes, as in the fragmented generic points of cheerleading strategic plans and the dreaded mission statement:

> ★ ***Accelerate The Introduction Of New Products !!!***
> ★ ***Accelerate Revenue Recognition !!!***

Better to say *who* will accelerate, and *what, how, when,* and *where* they will accelerate. An effective methodology for making such statements is the *sentence*, with subjects and predicates, nouns and verbs, agents and their effects. Identifying specific agents of action may also eventually help forensic accountants and prosecutors target those responsible for excessively accelerated recognition of revenue.

In presentations of plans, schemes, and strategies, bullet outlines get all confused even about simple, one-way causal models. Here, from the *Harvard Business Review*, a deep analysis of bullets for business plans:

> Bullets leave critical assumptions about how the business works unstated. Consider these major objectives from a standard five-year strategic plan
>
> - Increase market share by 25%.
> - Increase profits by 30%.
> - Increase new-product introductions to ten a year.
>
> Implicit in this plan is a complex but unexplained vision of the organization, the market, and the customer. However, we cannot extrapolate that vision from the bullet list. The plan does not tell us how these objectives tie together and, in fact, many radically different strategies could be represented by these three simple points. Does improved marketing increase market share, which results in increased profits (perhaps from economies of scale), thus providing funds for increased new-product development?
>
> Market share ⟶ Profits ⟶ New-product development
>
> Or maybe new-product development will result in both increased profits and market share at once:
>
> New-product development ⟶ Market share
> New-product development ⟶ Profits
>
> Alternatively, perhaps windfall profits will let us just buy market share by stepping up advertising and new-product development:
>
> Profits ⟶ New-product development ⟶ Market share.[3]

[3] Gordon Shaw, Robert Brown, Philip Bromiley, "Strategic Stories: How 3M is Rewriting Business Planning," *Harvard Business Review*, 76 (May-June, 1998), 44.

It follows that more complex and realistic multivariate causal models are way over the head of the simplistic bullet-list format.

To identify and critique assertions of effects without causes, consumers of presentations should sketch out the causal models lurking within the analysis, assess how the evidence links up to the theories, draw diagrams with arrows, label alleged causes and effects, contemplate the meaning of arrows, and do what the presenter should have done in the first place.

Cherry-Picking, Evidence Selection, Culled Data

THE most widespread and serious obstacle to learning the truth from an evidence-based report is *cherry-picking,* as presenters pick and choose, select and reveal only the evidence that advances their point of view. "It is a principle that shines impartially on the just and unjust that once you have a point of view all history will back you up."[4]

Few presenters are saintly enough to provide their audience with competing explanations, contrary evidence, or an accounting of the full pool of evidence tapped to construct the presentation. But thoughtful presenters might at least privately test their analysis: *Has evidence been filtered or culled in a biased manner? Do the findings grow from evidence or from evidence selection? Would the findings survive the scrutiny of a skeptic or investigator of research fraud? What would Richard Feynman think?* Such questions may help presenters get it right before enthusiasm for their own work takes over. To avoid being fooled, consumers of presentations must ask these questions as well.

It is idle, however, for skeptics to claim that the evidence presented in a report has been selected. Of course there is more evidence than that published. The key issue is whether evidence selection has compromised the true account—if we only knew what it was—of the underlying data.

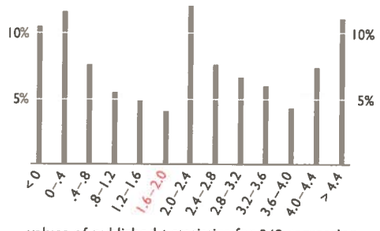
A CLEAR sign of cherry-picking is that a report appears *too good to be true,* provoking consumers of the report to mutter "*It's more complicated than that.*" But unless the back-office operations in preparing a report are known, it is difficult to identify the specific effects of evidence selection for any *single* report. A *series* of reports, however, can decisively reveal corrupt practices. It is perhaps merely a happy coincidence when the reported quarterly earnings for one corporation exceed its forecasted earnings by exactly one penny; but if this miracle occurs in 20% of all corporations, it is a sign of systematic manipulation of financial data.[5]

In medical research, too often the *first published study* testing a new treatment provides the *strongest evidence that will ever be found* for that treatment. As better controlled studies—less vulnerable to the enthusiasms of researchers and their sponsors—are then conducted, the treatment's reported efficacy declines. Years after the initial study, as the Evidence Decay Cycle plays out, sometimes the only remaining issue is whether the treatment is in fact harmful. Lack of disciplined comparisons, which gives free play to cherry-picking, erodes the credibility of early reports of enthusiasts, no matter how authoritative their sales pitch:

> One day when I was a junior medical student, a very important Boston surgeon visited the school and delivered a great treatise on a large number of patients who had undergone successful operations for vascular reconstruction. At the end of the lecture, a young student at the back of the room timidly asked, "Do you have any controls?" Well, the great surgeon drew himself up to his full height, hit the desk, and said, "Do you mean did I not operate on half of the patients?" The hall grew very quiet then. The voice at the back of the room very hesitantly replied, "Yes, that's what I had in mind." Then the visitor's fist really came down as he thundered, "Of course not. That would have doomed half of them to their death." God, it was quiet then, and one could scarcely hear the small voice ask, "Which half?"[6]

[4] Van Wyck Brooks, *America's Coming-of-Age* (New York, 1915), 20. Or, as Nero Wolfe remarks, "Once the fabric is woven it may be embellished at will." Rex Stout, *The Golden Spiders* (New York, 1955), 85.

[5] The distribution of published test statistics over a *series* of papers may reveal culling of evidence. Consider a research issue between those who believe there is no relationship between 2 variables (call this hypothesis H_0) and those who believe there is a relationship (H_A). Advocates of H_0 and H_A gather data, run many multiple regressions, and examine the *t*-statistics on the relevant regression coefficients. Both schools of advocates seek to publish decisive results; and both seek to avoid ambiguous results. Such advocacy may lead to evidence selection, resulting in a peculiar distribution of all published test statistics: a heaping of results that strongly favor *either* H_0 *or* H_A, and *fewer* results in the dreaded *Zone of Boredom, Ambiguity, and Unpublishability*, the zone of non-results. For *t*-tests on regression coefficients, the ZBAU values fall between 1.6 to 2.0, where the choice between H_0 and H_A is a close call. For my book *Political Control of the Economy* (Princeton, 1978), I compiled the distribution of the 248 published *t*-statistics from all 17 of the then-published studies of election year macroeconomic conditions and the U.S. national vote for the political party of the incumbent president. Both the H_0 and the H_A heapings, as well as the *Zone of Boredom, Ambiguity, and Unpublishability*, are seen below in the distribution of *published* *t*-statistics:



values of published *t*-statistics for 248 regression coefficients in all 17 published studies of economic conditions and election outcomes

THOMAS CHALMERS, a founder of evidence-based medicine, repeatedly demonstrated that the more susceptible a research design is to evidence selection and bias, the more enthusiastic the evidence becomes for favored treatments. For example, Chalmers and colleagues examined 53 published reports evaluating a surgical procedure, a portcaval shunt for esophageal bleeding.[7] All studies were rated on (1) *enthusiasm* of the findings for the surgery, (2) *quality of the research design* (good design = random assignment of patients to treatment or control groups; bad = treatment group not compared with any proper control). The gold standard of research designs is the randomized controlled trial (RCT), which assigns patients randomly to the treatment or the control group (assuring within chance limits that both groups are identical in all respects, known and unknown, thereby avoiding, for example, selection of more promising patients to favored treatments). Of the 53 published studies, only 6 were well-designed (RCT). And their findings were clear: none of the 6 well-designed studies were markedly enthusiastic about the operation:

Quality of research design versus degree of investigator enthusiasm for the portcaval shunt surgical procedure, 53 published studies

	MARKED ENTHUSIASM	MODERATE ENTHUSIASM	NO ENTHUSIASM
RESULTS OF 6 WELL-DESIGNED (RCT) STUDIES:	0	3	3
RESULTS OF 47 POORLY DESIGNED STUDIES:	34	10	3

In contrast, for 47 studies lacking valid controls, 34 expressed marked enthusiasm for the surgery. Thus 72% (34 of 47) of the poorly controlled studies got it wrong, endorsing a surgical procedure unwarranted by the RCT gold standard. This link between lousy research designs and wrongly enthusiastic reports has been replicated again and again for all too many drugs and surgical procedures (some eventually abandoned thanks to the meta-analysis of evidence similar to the table above).

Loosely designed studies allow the underlying medical reality to be filtered and cherry-picked so as to reliably produce unreliable evidence for favored treatments. Discovery of this persistent bias led to regulatory and scientific standards requiring that research on medical treatments use randomized controlled trials. These requirements, alas, have in turn fostered imaginative new methods for cherry-picking the data of medical research in favor of good news.[8]

[6] Dr. E. E. Peacock, Jr., University of Arizona College of Medicine; quoted in *Medical World News* (September 1, 1972), 45.

[7] N. D. Grace, H. Muench, T. C. Chalmers, "The Present Status of Shunts for Portal Hypertension in Cirrhosis," *Gastroenterology* 50 (1966), 684–691; table shown here as updated by Chalmers in John P. Gilbert, Richard J. Light, and Frederick Mosteller, "How Well Do Social Innovations Work?" in Judith Tanur, *et al.*, *Statistics: A Guide to the Unknown* (San Francisco, 1978), 135.

[8] An-Wen Chan, Asbjørn Hróbjartsson, Mette T. Haahr, Peter C. Gøtzsche, Douglas G. Altman, "Empirical Evidence for Selective Reporting of Outcomes in Randomized Trials: Comparison of Protocols to Published Articles," *Journal of the American Medical Association*, 291 (May 26, 2004), 2457–2465.

Controlled trials are *prospective:* a possible cause-effect relationship is identified, an intervention made, future outcomes observed. In contrast, evidence for many phenomena—weather, politics, geology, economics, art history, business—largely comes from *retrospective*, nonexperimental observation. In such after-the-fact studies, researchers and presenters have a good many opportunities to decide *what counts as relevant evidence,* which are also terrific opportunities for cherry-picking.

Consider a standard methodology in economics, finance, and political economy: explanations are empirically generated or perhaps evaluated by means of fitted models. Such multiple regression analyses of historical data seek to account for the past and to predict the future. In theory, perfectly reasonable; in practice, these efforts may be compromised by a certain slackness of theory and method:

(1) *Imprecise theories* Some theories suggest what variables might be related. But the theories tend to be vague and broad, hinting at perhaps 5 to 10 relevant effects, and 5 to 100 candidate causal variables.

(2) *Many "notions"* Researchers hold a variety of sub-theoretical ideas, or notions, that are employed in the course of data analysis: trying out variables not explicitly justified by the theory; excluding part of the data or mixing in *ad hoc* dummy variables to re-estimate the model; taking logarithms and other transforms; fitting innumerable lag structures to time-series. Researchers may have 5 to 100 notion options available.

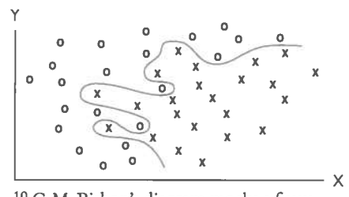
(3) *Many different operational measures for the same concept* Consider the variety of plausible empirical measures of economic growth, social status, cultural norms, educational achievement, political competition.

(4) *Data slack* In conducting data analysis, researchers decide many details: treatment of missing data, reconciliation of discrepant sources, construction of classifications, choice of the beginning/ending points in time-series (a notorious cheat in financial data), choice of category cut-points, and so on. Not all these decisions are necessarily made innocent of the favored result, and a few small tilts in the same direction soon add up to a finding.[9]

All told, many plausible models result. For k explanatory variables, there are $2^k - 1$ possible fitted models, then multiplied by notions and on through the rest of the slack. Routinely 10^4 to 10^7 fitted models are available; all can be quickly computed and sorted over. These models are not independent and many look pretty much alike, but which few to publish from millions of possibilities? This latitude for evidence selection makes it difficult to distinguish between reliable findings and cherry-pickings. Now and then it may matter.

Such model-searching might find something new and true. Or merely something brittle and over-fitted, a model that collapses when used for prediction.[10] Found models (possibly from among millions searched) must be replicated afresh on innocent data—for how can models produced

[9] "Exercising the right of occasional suppression and slight modification, it is truly absurd to see how plastic a limited number of observations become in the hands of men with preconceived ideas." Francis Galton, *Meteorographica* (London, 1863), 5.



[10] C.M. Bishop's diagram mocks a fussy over-fitted model (the red line) wandering around the data space picking up every little piece of stray variation. Built on the quicksand of idiosyncratic and random variation rather than the rock of predictive causal theory, over-fitted models shrink when applied to new data. Indeed, the decline in explained variance for a found model compared with that model applied to new data is appropriately called, in the jargon of model-building, "shrinkage".

by massive searches then be confirmed by reference to the same material? This circularity of stale testing of cherry-picked data-generated models leaves researchers and their findings no more secure than Ignorance's justification for hope in *The Pilgrim's Progress*:

> Ignorance: But my heart and life agree together, and therefore my hope is well grounded.
>
> Christian: Who told thee that thy heart and life agree together?
>
> Ignorance: My heart tells me so.[11]

[11] John Bunyan, *The Pilgrim's Progress* (1678), chapter 15.

Credible explanations grow from the combined testimony of 3 more or less independent, mutually reinforcing sources—explanatory theory, empirical evidence, and rejection of competing alternative explanations. Cherry-picking dilutes and confounds these 3 sources into the wishful circular thinking and just-so stories of Ignorance.

BETWEEN the initial data collection and the final published report falls the shadow of the *evidence reduction, construction, and representation process:* data are selected, sorted, edited, summarized, massaged, and arranged into published graphs, diagrams, images, charts, tables, numbers, words. In this sequence of representation, a report represents some data which represents the physical world:

raw data: observations, measurements → evidence reduction, construction, and representation → the report or presentation: findings represented by graphs, tables, diagrams, images, numbers, words

This process of evidence construction and representation, though not a black box but certainly a gray area, consists of all the decisions that *cause* the published findings of a report. These decisions are made, to varying degrees, both in the spirit of doing analytical detective work to discover what is going on *and* in the spirit of advancing a favored point of view. Thus the integrity of a report depends in part on the *integrity of the process* of evidence construction; alert consumers of a report must seek some kind of assurance that the process was sensible and honest.[12]

Given the persistent threat of cherry-picking and aggressive advocacy, consumers of reports and presentations might well ask: *Do the report's findings grow from the evidence or from the process of evidence construction? Would that process survive the scrutiny of a research audit? Does the presenter have a reputation for cherry-picking? Is the particular field of inquiry notorious for advocacy and evidence corruption (investment analysis, land development, new drug research, sales reports)? Are the findings too good to be true? Have the report's findings been independently replicated? How much does the decision to be made depend on the evidence in the report at hand? Who paid for the work?*

[12] John Ioannidis, "Why Most Published Research Findings Are False," *Public Library of Science*, 2 (August 2005), e124, a paper with a distinctly provocative title, suggests that in medical research: "A research finding is less likely to be true when the studies conducted in a field are smaller; when effect sizes are smaller; when there is a greater number and lesser preselection of tested relationships; where there is greater flexibility in designs, definitions, outcomes, and analytical modes; when there is greater financial and other interest and prejudice; and when more teams are involved in a scientific field in chase of statistical significance." Ioannidis's collection of references provides a summary of the considerable evidence on evidence corruption.

Punning, Overreaching, and Economisting

The book *Painting Outside the Lines: Patterns of Creativity in Modern Art* by David Galenson begins with an intellectual history of the work, disarmingly written in the first-person singular with 35 self-references in the first 2 pages. At right, the second page of the introduction. In the first paragraph here, *value* clearly refers not to merits but to *prices* of paintings, as art dealers, market transactions, and results of art auctions are mentioned. Described is a 2-variable study: the *price* of a painting in relation to the *age* of the artist when the painting was made. Fair enough, but why should anyone care about this correlation? Thus the key problem of the preface: can what might appear to be dustbowl empiricism be parlayed into an interesting book? And, if interesting, true?

Subtle shifts in language broaden the study's conceptual scope, although the underlying evidence remains unchanged. The word *value* migrates into the comparative, as high-priced paintings are described as the "most valuable work" of an artist. This comes close to a pun, since the dictionary meanings of *valuable* are (1) monetary worth and (2) meritorious, admirable, esteemed qualities. The paragraph's final sentence completes the punning equation, as a rhetorical maneuver turns auction prices into "*importance* of artistic work" via the intermediate term of "most valuable work."

The intermediate link "most valuable work" now blithely converts auction prices into a measure of "best work" of an artist. Views on this matter differ. "The present commercialization of the art world, at its top end, is a cultural obscenity," said Robert Hughes, an expert on art and art markets, "When you have the super-rich paying for an immature Rose Period Picasso $104 million [in 2004], close to the GNP of some Caribbean or African states, something is rotten. Such gestures do no honor to art; they debase it making the desire for it pathological." Can rarely-auctioned paintings be understood as commodities? Perhaps the auction price for a painting also reflects previous prices (statistically adjusted for inflation), trends in economic conditions, varying fashions for artists, dealer and auction-house marketing, commission price-fixing, changes in the number of customers for pricey art, provenance of paintings, and gullibility of art-collecting *nouveau riche* adjusted for trendy. Variations in auction prices result in part from distinctive characteristics of art markets (see, for example, 2 recent books: Meryle Secrest, *Duveen: A Life in Art* and Christopher Mason, *The Art of the Steal: Inside the Sotheby's-Christie's Auction House Scandal*).

But wait: the book is also about the life cycle of *artistic productivity*, measured in part by . . . *auction prices*. There's more: auction prices help explain how *artists think and work*. Just imagine that. How does today's price at Christie's or Sotheby's explain how Cézanne "planned and executed his work" 120 years ago? Backward run the inferences until reels the mind.

Since my college course I had remained interested in modern art. In the spring of 1997, in talking with several art dealers, I was intrigued to learn that the value of a particular contemporary artist's work had declined over the course of his career. Wondering how common this was, I realized that I could find out systematically, in much the same way I had measured age effects in my earlier work: I could use market transactions—the results of recent auctions—to estimate the relationship between the value of an artist's paintings and the artist's age at the time of their execution. During the following summer I collected the appropriate data and made these estimates for the most prominent American painters of two generations, the Abstract Expressionists and their successors.

The results were startling. The most valuable work of Jackson Pollock, Mark Rothko, and the other Abstract Expressionists was almost invariably done late in their careers, but just the opposite was true for Jasper Johns, Robert Rauschenberg, and the other major painters of the next generation. The emphasis of my college course on the importance of the early work of Johns and Stella had therefore not been an accident, for the leading artists of their generation almost all produced their most valuable work at early ages.

Curious whether these results were unique to New York in the 1950s and 1960s, I then made a similar study of the careers of the great French painters who dominated the first century of modern art. To my surprise, I again found evidence of a shift over time. Modern painters in France born before 1850, including Manet and Cézanne, normally produced their most valuable work late in their careers, but the leading artists in the generations that followed, from Gauguin and van Gogh through Picasso and Braque, typically did their best work when they were much younger.

This book presents the results of these two studies of the relationship between age and productivity, together with my interpretation of the causes of those results and consideration of some of their consequences. In doing this research, I found the same excitement in identifying and explaining systematic patterns in the history of art that I have always found in doing similar work on problems in economic and social history. I was amazed how often the results of quantitative analysis, whether of auction prices or textbook illustrations, could lead to accurate predictions about how individual painters conceived of their enterprise as artists, and even about how they went about planning and executing their work. But I was also disappointed to discover how completely art historians have neglected quantitative approaches to their discipline. Such work can offer new insights into the history of modern art, and in so doing adds another dimension to existing work based on traditional approaches.

Above, David Galenson, *Painting Outside the Lines: Patterns of Creativity in Modern Art* (Cambridge, Massachusetts, 2001), xiv. Quotation at left: Robert Hughes, "A Bastion Against Cultural Obscenity," *The Guardian*, June 3, 2004.

Fischer's *Historians' Fallacies* describes the punning multiplicity of meaning as "*the fallacy of equivocation* . . . whenever a term is used in two or more senses within a single argument, so that a conclusion appears to follow when in fact it does not." Auction prices allegedly carry enormous information, relevant one way or another to all the following: (1) the most valuable, meaning both price and merit, work of an artist, (2) historical importance, (3) best artistic work, (4) artistic productivity, (5) creativity, (6) how artists conceive their works, (7) how artists paint their works. Consider a thought experiment: *Do our theories about (1), (2), (3), (4), (5), (6), and (7) change when new data (say, in 2010-2020) for auction prices become available?* If yes, how and why should art history be rewritten? If no, what exactly is the relevance of auction prices for understanding "patterns of creativity in modern art"?

Concepts that explain everything explain nothing. Such concepts deny distinctions between cause and effect, theory and evidence, explanation and correlation. Here the explanatory meaning of "price" in economic history is as mushy as the meaning of "culture" in anthropology, which Clifford Geertz devastatingly describes as "so diffuse and all-embracing as to seem like an all-seasons explanation for anything human beings might contrive to do, imagine, say, be, or believe. . . . We were condemned, it seemed, to working with a logic and a language in which concept, cause, form, and outcome had the same name."

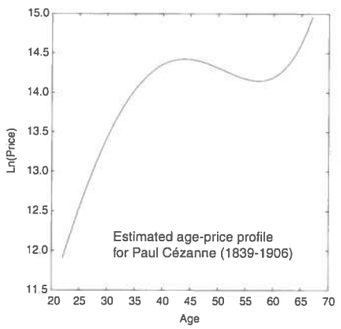
The artist's *age* at the time a painting was painted, the other key variable, also has a multiplicity of meanings. When Picasso was 26, it was *also* 1907, a magical year, the beginning of Cubism. Thus an artist's *age* is confounded with the *year* the painting was created and everything going on during that moment in art. Thus for some artists at some times, age is a *proxy* (statistical jargon for "pun") for art history.

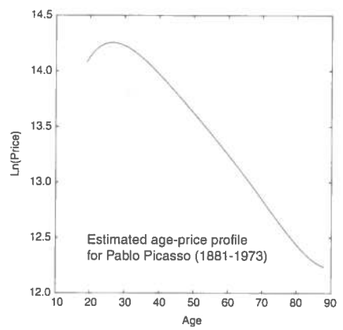
At least Professor Galenson is satisfied. "Systematic" analysis has produced "startling" findings, which create "surprise," "excitement," "new insights." He is "amazed how often [his] results lead to accurate predictions" although "also disappointed how completely art historians have neglected" his approach. The problem here is not the self-congratulation, but rather that *self-reported self-astonishment is presented as evidence for the credibility of one's own research.*

Slippery language, stupendous conclusions. This syndrome of overreaching is economically described by a new word: *economisting*, with accents on *con* and *mist:*

economisting (e **kon**′o **mist**′ing) 1. The act or process of converting limited evidence into grand claims by means of punning, multiplicity of meaning, and overreaching. 2. The belief or practice that empirical evidence can only confirm and never disconfirm a favored theory. 3. Conclusions that are theory-driven, not evidence-based. See also *confirmation bias, painting with a broad brush, Iraqi weapons of mass destruction, marketing, post-modern critical theory, German meaning of "mist".*

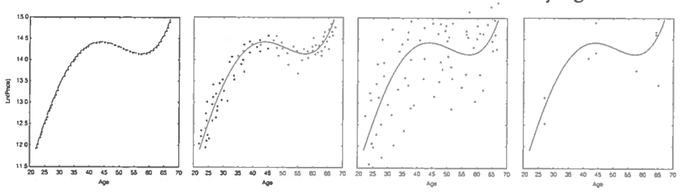
Quotations at right: David Hackett Fischer, *Historians' Fallacies* (New York, 1970), 274; Clifford Geertz, *Available Light: Anthropological Reflections on Philosophical Topics* (Princeton, 2000), 12–13.





David Galenson, *Painting Outside the Lines: Patterns of Creativity in Modern Art* (Cambridge, Massachusetts, 2001), 15, figures 2.1 and 2.2.

The puns in *Painting Outside the Lines* nearly make some empirically testable claims about the extraordinary explanatory powers of auction prices. What then is provided in the way of *evidence* about artists, paintings, and prices? For one thing, the book presents 15 data tables with 2,029 entries (artists' birthdates, deathdates, ages, and frequency of appearances in exhibitions and art history textbooks)—but *not a single auction price or price index or anything else that might measure an economic transaction for a particular painting.* Likewise for the text in the 265-page book: no prices. The book's 2 graphs show a vertical axis, Ln(Price) = the natural logarithm of prices, along with age/price curves for Cézanne and Picasso. These 2 tidy curves, without any actual data points, are the *only* quantitative evidence about auction prices in the book. Readers are not even told the number of paintings plotted graphically. The Cézanne age/price curve shown above are consistent with all sorts of underlying data:



What information is necessary to assess the credibility of the book's 2 graphs? Here is a checklist, familiar to students of research methods:

Number (n) of paintings in the analysis.
The data matrix (auction price and age of artist, for the n paintings).
Equations of the fitted models, and plotted curves accompanied by the data.
Quality of fit of the estimated models.
The substantive meaning, quantitatively expressed, of the estimated models.[13]

Thoroughly dequantified, *Painting Outside the Lines* provides *none* of this information. Elementary standards of statistical evidence are not met by the book's notable publisher (Harvard University Press) or the notable author (Professor of Economics, University of Chicago). The economisting puns are unsupported by the economisting evidence.

[13] The 2-dimensional space for these graphs can yield intriguing quantitative interpretations, a point missed by the fitted polynomial curves at left (which use t^0, t^1, t^2, t^3, t^4, t^5 terms to fit auction prices). Instead, consider an exponential model, where p = *auction price* and t = *age of the artist*:

$$p = ae^{bt}$$

Taking natural logarithms and then letting $c = \log_e a$ allows the model to be estimated by ordinary least squares (OLS). Conveniently, this is also the 2-space of the graphs above:

$$\log_e p = c + bt$$

The regression coefficient, b, has a useful substantive interpretation: in the model $p = ae^{bt}$, $b \times 100$ is equal to the percent increase in p per unit increase in t, if b is small (say, less than 0.25). The proportional increase in p, per unit increase in t, is

$$\frac{\frac{\Delta p}{p}}{\Delta t} = \frac{p_2 - p_1}{p_1} \quad (\text{since } \Delta t = t_2 - t_1 = 1)$$

$$= \frac{ae^{bt_2} - ae^{bt_1}}{ae^{bt_1}} = e^b - 1$$

$$= \left(1 + b + \frac{1}{2!}b^2 + \frac{1}{3!}b^3 + \ldots\right) - 1$$

by the series expansion of e^b. If b is small, then higher-order terms can be dropped

$$\approx (1 + b) - 1 = b$$

the OLS slope in the $\log_e$*price* by *age* space. Another possibility is to estimate elasticities by taking the logarithm of both variables. As insiders know, however, the resulting R^2 will be much smaller compared to letting 6 polynomial age terms bend and wander around to fit auction prices. What theory of markets—other than barefoot empiricism—specifies such a model? Higher powers of an artist's age make no *substantive* sense; at a vigorous 80, Picasso's age to the fifth is a stupefying 3,276,800,000. What then can the regression coefficient on t^5 mean: "A 100,000,000 quintic-year change in *artist age* explains a β_5 change in $\log_e$*prices*"?!

In REPORTS on quantitative work, frequent puns involve the language of mathematical statistics: *significance, confidence, maximum likelihood, bias, standard errors, optimal.* (Recall the "strict consensus of parsimonious trees with optimized activity patterns" in our earlier example of fitted cladograms.) In statistics, these words have clearly defined technical meanings. The everyday meanings of these words, however, resonate far beyond their narrow technical usage.

Statistical tests against the null hypothesis allow some researchers to make punning claims about the significance (everyday meaning) of their findings. Statistical significance (technical meaning) derives from the ridiculousness of the null hypothesis, sample size, fateful and usually false assumptions about independence of observations, assumptions about the sampling distribution under the null hypothesis, and, yes, size of the effect. Thoughtful researchers will report content-relevant measures of effects and not confound those measures with tests against the null hypothesis. Insiders already know this, but the punning problems persist in workaday research practice. Such puns identify mediocrities.

Puns from microeconomics and mathematical statistics—and, for that matter, quantum mechanics, evolutionary theory, fractals, chaos theory—claim unmerited credibility by trading on the authority and sometimes the jargon of the original narrow technical achievement. Puns enable *overreaching*, as previously bright ideas sprawl, grow mushy, and collapse into vague metaphors when extended outside their original domain. Steven Weinberg describes the breakdown in logic when once-precise concepts overreach:

> Quantum mechanics has been variously cited as giving support to mysticism, or free will, or the decline of quantitative rationality. Now, I would agree that anyone is entitled to draw any inspiration he or she can from quantum mechanics, or from anything else. This is what I meant when I wrote that I had nothing to say against the use of science as metaphor. But there is a difference between inspiration and implication, and in talking of the "telling cultural implications" of quantum mechanics, Professor Levine may be confusing the two. There is simply no way that any cultural consequences can be *implied* by quantum mechanics. It is true that quantum mechanics does "apply always and everywhere," but what applies is not a proverb about diverse points of view but a precise mathematical formalism, which among other things tells us that the difference between the predictions of quantum mechanics and pre-quantum classical mechanics, which is so important for the behavior of atoms, becomes negligible at the scale of human affairs.[14]

[14] Steven Weinberg, *Facing Up: Science and Its Cultural Adversaries* (Cambridge, Massachusetts, 2001), 156–157.

When a precise, narrowly focused technical idea becomes metaphor and sprawls globally, *its credibility must be earned afresh locally by means of specific evidence demonstrating the relevance and explanatory power of the idea in its new application.* It is not enough for presenters to make ever-bolder puns, as meaning drifts into duplicity. Something has to be explained.

Chartjunk: Content-Free Stuff Replaces Evidence

For consumers of presentations, gratuitous and cartoonish decoration of statistical graphics provides evidence about the presenter's integrity and statistical skills: little integrity, no statistical skills. Imagine the quality of analysis behind this chart: a fanciful 3-fold change in revenue growth is depicted by a 7-fold change in bar *area* and an immense change in the apparent *volume* of the guy in the track suit. And who would believe in revenue growth projections for some 4 years?

For cynical or malicious presenters, chartjunk decoration reflects their contempt for evidence and for their audience. Chartjunk flows from the premise that audiences can be charmed, distracted, or fooled by means of content-free misdirection: garish colors, designer colors, corny clip-art, generic decoration, phony dimensionality. For decoration, it sure is ugly.

Audience members at a presentation featuring chartjunk rather than evidence should ask themselves "Is this the quality of analysis that we are relying on to understand a problem or to make a decision? Why should we trust this presenter? Just how high can the presenter count? Does the presenter think we're fools? Why are we having this meeting?"

Microsoft Excel and PowerPoint produce, ineptly, many of the data graphics and tables used in presentations today. Filled with chartjunk, the default graph templates in these programs are useful for constructing deceptive investment and weight-loss pitches. Excel chartjunk can sometimes be finessed by skilled users; PowerPoint graph templates are broken beyond repair. For preparing data presentations other than ads in tabloid newspapers, a professional statistical graphics program is essential.

ALONG with the *chartjunk of garish decoration*, there is also the *chartjunk of graph bureaucracy*: useless or optically active grids, boxes and frames around graphs, redundant representations of data, cross-hatched bars. In this spreadsheet assessing the danger to the Columbia space shuttle, the most prominent visual activities are the vast empty framing areas and the grid prisons that surround the unexplained and unreadable numbers.

Very little chartjunk appears in the sports, weather, and financial tables in newspapers, or in the tables and graphs published in major scientific journals—since the content is too important and too complex for fooling around with chartjunk.[15]

[15] On chartjunk, see Edward Tufte, *The Visual Display of Quantitative Information* (Cheshire, Connecticut, 1983, 2001), chapter 5.

Results of Impact Analysis for particle size = 20" x 10" x 6"

XT	YT	ZT	VMAX (ft/sec.)	VX (ft/sec.)	VY (ft/sec.)	VZ (ft/sec.)	IMPANG (degrees)
1755	193	625	690	682	104	20	9.0
1759	194	630	689	680	107	25	9.4
1744	190	637	693	683	107	36	8.7
1755	191	641	698	689	107	41	7.8
1800	197	648	702	693	105	46	8.8
1747	190	626	686	677	104	21	7.0
1769	192	629	682	674	105	23	7.1
1751	188	637	685	676	105	35	10.4
1754	188	641	690	681	104	40	7.8
1754	187	644	694	684	103	44	6.6
1755	197	627	693	684	107	23	11.9
1748	195	630	691	682	107	27	13.3
1756	194	638	699	689	109	37	8.9
1806	202	645	712	703	109	42	11.3
1788	199	647	711	701	109	46	10.4
1762	200	627	700	691	109	24	21.5
1833	211	633	707	698	110	28	9.6
1802	204	641	713	703	110	38	12.8
1790	202	644	711	702	110	42	11.3
1781	200	647	712	703	108	46	11.1
1744	186	625	683	675	102	18	6.5
1718	181	627	673	665	101	22	6.0
1742	184	636	653	645	98	30	2.0
1652	169	635	635	627	96	32	0.4
1593	159	634	611	603	92	34	2.0
1786	198	621	705	697	104	15	7.5
1799	201	624	702	694	105	18	7.7
1758	194	624	691	683	104	20	9.1
1830	210	617	723	715	106	12	5.4
1799	205	620	710	702	106	15	7.9
1790	202	623	707	699	106	17	8.1
1762	198	625	694	686	107	21	11.8
1788	196	620	705	697	102	14	7.0
1798	198	623	698	691	103	17	7.2
1755	191	624	687	679	103	19	6.8
2023	238	615	762	755	103	7	1.1
1830	210	617	723	715	106	12	5.4

STS-107 Debris Impacting Orbiter Wing

BOEING

When Evidence is Mediated and Marketed: Pitching Out Corrupts Within

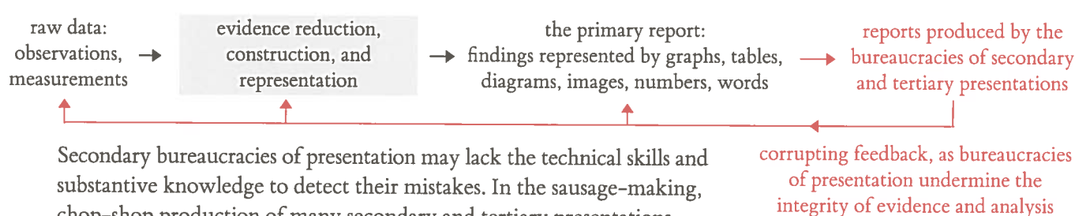
EVIDENCE-BASED reports are repackaged and marketed by *bureaucracies of secondary presentations*: public relations, advertising, programs for public outreach, schoolbook publishing, journalism, and the vast government Ministries of Propaganda. Soon enough, tertiary presentations pitch recaps of opinions about a summary of some evidence somewhere.

In repackagings, a persistent *rage to conclude* denies the implications, complexities, and uncertainties of primary evidence.[16] A strong selection bias typically operates: news wins out over olds, recency rather than the quality of evidence decides the relevance of evidence.

Secondary re-presentations have much larger audiences than primary reports. Rarely read, the original report blurs and fades away. A handful of technical reviewers might examine the complete set of evidence for a new drug; possibly 500 people read medical journal articles about the drug and 5,000 peruse medical abstracts; 500,000 might see news reports and millions advertisements for the drug, as the original evidence has ultimately passed through 3 or 4 repackagings on the way to market. Or consider textbooks: a successful college text is assigned to 200,000 students, some of whom read it; the primary works summarized by the textbook are read by a few researchers. For reports of government commissions, for *each* reader of the original, there are perhaps 100,000 readers of mediated secondary versions. Sometimes, actually reading a government report constitutes original research.

Substantial resources are devoted to repackaging: pharmaceutical companies spend more on marketing than on drug research, schoolbook publishers more on lobbying textbook selection committees than on writing books, financial firms more on promoting investment products than on discovering them.

In repackaging and commodifying evidence-based material for wider distribution, the bureaucracies of secondary presentation get their whack at the primary report: they edit, clarify, interpret, summarize, simplify, over-simplify, spin, tart up, mess up. And they make errors. If you worry about evidence corruption in primary reports, secondary presentations will give you a lot more to worry about. Repackaging adds its own special interpretative filter to the critical process of learning from evidence:



Secondary bureaucracies of presentation may lack the technical skills and substantive knowledge to detect their mistakes. In the sausage-making, chop-shop production of many secondary and tertiary presentations, absent are methods that routinely help enforce the intellectual quality and integrity in primary work: external review and final approval by content experts, professional standards of evidence, skeptical intelligence. And then, to impede direct communication between originators and audiences, bureaucracies of secondary presentation may limit access to the primary report through copyrights, inconvenient or costly subscriptions, and overreaching claims of corporate privilege or government secrecy.

[16] Flaubert's phrase "*la rage de vouloir conclure*" occurs in his letter to Mlle Leroyer de Chantepie, October 23, 1863, here translated: "The rage for wanting to conclude is one of the most deadly and most fruitless manias to befall humanity. Each religion and each philosophy has pretended to have God to itself, to measure the infinite, and to know the recipe for happiness. What arrogance and what nonsense! I see, to the contrary, that the greatest geniuses and the greatest works have never concluded." Gustave Flaubert, *Correspondance* (Paris, 1929), vol. v, 111.

PRODUCERS of primary reports may console themselves about distortions in mediated versions of their work by recalling the words of the noted marketeer P. T. Barnum: "Without publicity a terrible thing happens—nothing." Evidence cannot become relevant if no one knows about it.

How can creators of evidence-based reports defend the integrity of their work against repackaging mischief? To start with, primary reports should be inexpensively and directly available (internet, self-publishing, leaks to journalists), short-circuiting the bureaucracies of secondary presentation. Creators of original material should never surrender rights to their work. They should prepare their own secondary reports to replace repackagings. They should police secondary mediated versions of their work, and turn the mistakes of the easily mocked pitch culture into notorious examples.

And *consumers* of evidence should stay reasonably close to primary sources and to evidence-interpreters who provide honest unbiased readings. There's almost always something to learn from a first-rate primary source (for example, that secondary versions incorrectly interpret the primary). For consumers, an indicator of an untrustworthy presentation bureaucracy is its denial of access to primary evidence (for example, by requiring that all publications be pre-approved by the PR department). Another sign is that repackagings always manage, somehow, to support a predetermined line. Wise consumers should keep in mind that reports from an unbiased interpreter will not always agree with their own prior views. And that one's allies are not all that less likely to corrupt evidence than one's opponents.[17]

BY generating corrupt repackagings, an organization's bureaucracy of secondary presentations may come to corrupt the integrity of work *within* the organization. Compromised external communications promote compromised internal communications, as *pitching out corrupts within*. If a corporation distorts evidence presented to consumers, stockholders, and journalists, then it may soon lie to itself. Or, similarly, the chronic problem of government intelligence agencies: once the collection and selection of evidence starts to become fixed around a pre-determined policy line, intelligence agencies may become perpetually unintelligent, confused about the differences between detective work and marketing. Or, even in writing novels: "Cliché spreads inwards from the language of the book to its heart. Cliché always does."[18] Nowadays the common tool for pitching out—and corrupting evidence within—is PowerPoint.

[17] A good guide for assessing the quality and the credibility of nonfiction reports is Sarah Harrison Smith, *The Fact Checker's Bible: A Guide to Getting It Right* (New York, 2004).

[18] Martin Amis, *The War Against Cliché* (New York, 2001), p. 137.

The English language . . . becomes ugly and inaccurate because our thoughts are foolish, but the slovenliness of our language makes it easier for us to have foolish thoughts.

George Orwell, "Politics and the English Language"

For a successful technology, reality must take precedence over public relations, for Nature cannot be fooled.

Richard P. Feynman, *"What Do You Care What Other People Think?"*

And not waving but drowning.

Stevie Smith, poem, "Not Waving But Drowning"

Sweet songs never last too long on broken radios.

John Prine, "Sam Stone"